HEALTHCONNECT ANALYTICS REPORT

Week 5: Exploratory Analysis, KPI Development & Business Insights

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PROJECT	HealthConnect Appointment Analysis	**DATE**	September 2026

1. EXECUTIVE SUMMARY & DATA PREPARATION AUDIT

This report evaluates appointment attendance patterns, operational friction points, and risk drivers across the HealthConnect Clinic dataset. The objective is to identify data-driven interventions that reduce patient no-shows, optimize practitioner capacity, and mitigate revenue leakage.

DATA PREPARATION & QUALITY OBSERVATIONS

To protect data integrity, the original dataset (`HealthConnect_Appointment_Data.csv`) was preserved in read-only format. A working dataset (`HealthConnect_Week5_Cleaned_Data.csv`) was generated following strict audit and transformation protocols:

- **Duplicate Inspection:** Audited unique `appointment_id` keys; 0 duplicate records were detected.
- **Missing Value Imputation:** Missing continuous features (`waiting_time_minutes`, `distance_to_clinic_km`, `booking_lead_days`, `age`) were imputed using feature medians to maintain robustness against skewed distributions. Missing categorical indicators (`reminder_channel`, `gender`) were populated with `'None'` and `'Unknown'` placeholders.
- **Outlier & Inconsistency Resolution:** Negative values in `booking_lead_days` were clipped to `0` (representing same-day bookings). Text columns were whitespace-stripped and lowercased to prevent sparse encoding artifacts.
- **Feature Engineering:** Formulated a zero-division safe `prior_no_show_ratio = previous_no_shows / (previous_appointments + 1)` and mapped binary target outcome labels (`1` = No-Show, `0` = Attended).

2. KEY PERFORMANCE INDICATORS (KPIS)

KPI NAME	FORMULA / DEFINITION	VALUE	BUSINESS QUESTION & IMPACT
1. Overall No-Show Rate	(Total No-Shows / Total Scheduled Appointments) × 100	**25.00%**	*What is our baseline capacity loss?* Establishes the clinic-wide non-attendance rate driving unutilized provider time.
2. Long Lead-Time Default Rate	(No-Shows [>14 Lead Days] / Total Appointments [>14 Lead Days]) × 100	**42.30%**	*How does advance booking impact commitment?* Quantifies patient intent decay when appointments are booked weeks in advance.
3. Unreminded No-Show Rate	(No-Shows [Channel = 'None'] / Total Unreminded Bookings) × 100	**38.80%**	*What is the cost of communication failure?* Measures attendance drop-offs resulting from missing outreach triggers.
4. Recurrent Offender Ratio	(No-Shows [≥2 Prior No-Shows] / Total Patients [≥2 Prior No-Shows]) × 100	**56.10%**	*Are past behaviors persistent?* Identifies high-risk, chronic non-

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			attenders requiring specialized intervention strategies.

3. TOP 5 BUSINESS INSIGHTS

1. **Lead-Time Decay Effect:** Attendance probability exhibits a steep linear degradation as booking lead time increases. Appointments booked >14 days in advance account for over 40% of all clinic no-shows, whereas same-week bookings (0–3 days) maintain an attendance rate above 85%.
2. **Outreach Channel Efficacy:** Conversational, two-way mobile channels (**WhatsApp** and **SMS**) yield the lowest default rates (14.2% and 18.5% respectively). Email and unreminded workflows (38.8% default) fail to engage patients effectively.
3. **Behavioral Recurrence:** Historical patient behavior is the single strongest predictor of future non-attendance. Patients with 2 or more prior no-shows default on subsequent appointments over 56% of the time.
4. **Geographic Frictional Barrier:** Patients residing >15 km from the facility demonstrate significant attendance variance. Distance acts as an operational friction point, compounded by local public transit constraints and extended travel times.
5. **Specialty Capacity Leakage:** High-demand departments (e.g., Cardiology and General Practice) experience severe operational inefficiency due to unutilized slots that cannot be filled on short notice.

4. STRATEGIC RECOMMENDATIONS

- **Automated Re-Engagement Sequences:** Deploy automated WhatsApp confirmation triggers at 7 days, 48 hours, and 24 hours prior to appointment times for all bookings with lead times >7 days.
- **Calibrated Overbooking Margins:** Implement a dynamic 10%–15% overbooking buffer specifically during peak long-lead time slots and high-risk department hours.
- **Tele-Health Triage for Distant Patients:** Automatically offer virtual/tele-health consultations to patients residing >15 km from the clinic to remove geographic travel barriers.
- **Deposit / Priority System for Recurrent Offenders:** Require patients with ≥2 prior no-shows to confirm attendance via a mandatory 24-hour phone check-in before locking in peak morning slots.

5. PROJECT LIMITATIONS

- **Socioeconomic Variables:** The dataset lacks economic indicators (e.g., income brackets, insurance status, transit access) that influence patient mobility.
- **Environmental & Seasonal Factors:** Analysis does not account for weather disruptions, public holidays, or time-of-day clinic congestion patterns.
- **Uncaptured Cancellations:** Patient-initiated early cancellations were not always distinct from unexcused no-shows in historical records.